

SAYOOJ.K.P

Mechanical Engineer | Industrial & Manufacturing | QA/QC | Technical Documentation | AI/LLM Evaluation & Prompt Engineering | SAP MM | BPCL Refinery Experience

B. Tech Mechanical Engineering

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Mechanical Engineer with 6+ years of experience across industrial engineering, refinery operations, plant maintenance, QA/QC, technical documentation, manufacturing content validation, and AI/LLM evaluation. Experienced in BPCL refinery operations, SAP MM, technical quality review, engineering content assessment, AI/LLM evaluation, and prompt engineering for mechanical and manufacturing use cases. Strong ability to combine mechanical engineering knowledge, analytical problem-solving, technical documentation, and AI technologies to deliver accurate and reliable outcomes.

CAREER HIGHLIGHTS

- 6+ years of professional experience across Mechanical Engineering, Plant Maintenance, QA/QC, Technical Content Development, and AI Evaluation.
- Graduate Engineer Trainee at BPCL Kochi Refinery, supporting refinery maintenance, shutdown, and startup operations.
- Manufacturing Domain Expert and AI Generalist Expert at Mercor, contributing to AI model evaluation and engineering content validation.
- University Rank Holder in B.Tech Mechanical Engineering with First Class Honours (CGPA: 8.16).
- Experienced in SAP MM, technical documentation, accessibility-focused content/ALT text, and engineering content review.

PROFESSIONAL EXPERIENCE

AI Generalist Expert – Mercor

Contract | Remote | 2026

- Evaluated AI-generated responses for factual accuracy, reasoning quality, instruction adherence, contextual understanding, and communication.
- Designed and tested prompts for mechanical engineering and technical scenarios to evaluate AI reasoning, technical accuracy, and instruction following.
- Provided structured feedback to assess and improve LLM response quality.
- Reviewed AI outputs for consistency, logical reasoning, and adherence to task-specific requirements.

Manufacturing Domain Expert – Mercor

Contract | Remote | 2026

- Evaluated AI-generated mechanical engineering and manufacturing responses for technical accuracy, industrial feasibility, and process reliability.
- Validated AI outputs against engineering principles and real-world industrial practices.
- Reviewed AI-generated content covering maintenance engineering, production systems, reliability, quality control, thermodynamics, and fluid mechanics.

Senior Subject Matter Expert - Mech, QA/QC Reviewer, ALT text - Innovalance Learning systems Pvt.Ltd

Sept 2022 – Sept 2025

- Performed technical quality checks, error analysis, and approval processes to ensure accuracy of mechanical engineering content.
- Reviewed and validated technical documentation and educational engineering content for accuracy, consistency, and quality standards.
- Produced accessibility-compliant ALT text descriptions for engineering diagrams, technical illustrations, and educational content to support inclusive digital learning.
- Applied mechanical engineering knowledge across thermodynamics, fluid mechanics, heat transfer, machine design, and related subjects.
- Used AI-enabled tools to support content validation and improve workflow efficiency and turnaround time.
- Ensured deliverables met industry, academic and client standards while maintaining high productivity.

Subject Matter Expert (SME) - Relevel by Unacademy

May 2022 – Aug 2022

- Delivered domain support for mechanical concepts, improving learner understanding and question resolution efficiency.
- Simplified technical concepts to improve learner experience and reduce repeat query volume by 32%.

Translation and Transcription (Robot training Malayalam) - Exonerate, Kochi

Freelancer - Jan 2022 – Apr 2022

- Translated and transcribed 6000+ Malayalam voice/text data entries with 99% accuracy, supporting AI/robotic voice-training model development.

Graduate Engineer Trainee - Bharat Petroleum Corporation Limited (BPCL), Kochi

Jan 2020 – Oct 2021

- Supported shutdown and startup activities of plant operations including gas turbines, boilers, cooling towers, and nitrogen plant operations.
- Participated in preventive and corrective maintenance activities for critical refinery equipment.
- Worked with SAP MM processes including material requisition and resource tracking.
- Assisted in shutdown planning and execution while maintaining strict safety and operational standards.
- Coordinated with multidisciplinary teams to ensure timely completion of maintenance activities.
- Assisted supervisors in supervising contractors, field safety compliance, and documentation.

Mechanical Engineer - Green Hub Engineering Contractors, Kochi

Dec 2017 – Nov 2019

- Supervised on-site mechanical work and contractor activities to maintain strict compliance with safety and equipment specifications.
- Monitored work progress to ensure adherence to specifications and safety requirements.

Subject Matter Expert - Chegg India PVT.LTD (Part Time/Remote)

Aug 2017 – Dec 2021

- Solved advanced mechanical engineering problems with high solution accuracy and on-time delivery.
- Leveraged analytical thinking and domain expertise to ensure precision, timely delivery, and conceptual clarity in solutions.

EDUCATION

B. Tech – Mechanical Engineering (First Class with Honours) – 2017

Nehru College of Engineering & Research Centre — University of Calicut

CGPA: 8.16 — University Rank Holder

CORE COMPETENCIES

- Technical Content Review & Validation • Manufacturing Domain Expertise • Reliability Engineering • Plant Operations • Technical Inspection • Troubleshooting • Project Execution • Preventive Maintenance • Predictive Maintenance • Asset Management • Equipment Installation & Testing • AI/LLM Evaluation • Prompt Engineering • Technical Documentation • Root Cause Analysis (RCA) • Accessibility-Focused Content / ALT Text • Engineering Content Review • SAP MM

PROFESSIONAL STRENGTHS

- Analytical & Problem-Solving • Technical Communication • Attention to Detail • Cross-Functional Collaboration • Adaptability & Learning • Organization & Time Management

TECHNICAL SKILLS

- Microsoft Office Suite (Excel, Word, PowerPoint) • SAP MM • Engineering AI Tools • MATLAB • AutoCAD • SolidWorks • Python Basics • C & C++ • Cyber Security Basics • Accessibility (ALT Text)

CERTIFICATIONS

- Rubric Academy & Prompt Academy – Fellowship, Mercor (2025) • SAP MM (Material Management) • Project Management Essentials (PMEC) • Lean Six Sigma White Belt • Six Sigma Yellow Belt • Seven QC Tools • Cyber Security Professional • Computer Programming in C & C++ • Positive Intervention and Situational Awareness Induction

PROJECTS AND SEMINAR

- Design & Fabrication of Concentric Tube Heat Exchanger Using Nanofluids - [Project](#)
- Experimental Investigation of Ethyl-Esterified Rice Bran Petrol Fuel Blend - [Project](#)
- Hydrogen–Ethanol Blending as an Alternative Fuel in SI Engines - [Seminar](#)

LANGUAGES

- English • Malayalam • Hindi • Tamil

HONORS & ACHIEVEMENTS

- University Rank Holder – Calicut University • IPATE Qualified • Scholarship – EdukateAKid

TRAININGS

- Indian Railway Palakkad MEMU Shed - [Implant Training \(1 Month\)](#)
- Prakash Industries Limited, Bangalore - [Industrial Visit](#)